Classic hedging is all you need

Classical versus learned option hedging under transaction costs

Replications of Sepp (2013) and Buehler et al. (2019), with a head-to-head comparison

DSC4107 Final Project

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Main body under 2000 words. Source code in Appendix B.
All results generated by the scripts in `submission/code/`.

Abstract

Deep hedging is sold as strictly freer than classical bands: given costs, a neural policy should beat a tuned delta schedule. Sepp (2013) and Buehler et al. (2019) attack the same friction from opposite ends, yet neither runs a fair head-to-head on one simulator with a classical baseline re-tuned per regime. We do. At the course's nominal 20 bp friction, re-tuned classical does not lose in any hedging sense that matters: Sharpe edges sit between about -0.09 and $+0.01$; a locked holdout confirms only $+0.0339$; and the learned terminal P&L buys mean with skew -0.70 against -0.01 and a 24% worse CVaR at 95%. Classic hedging is enough; deep is not a uniform upgrade. Above roughly 50 bp, large Sharpe gaps track P&L standard deviation rising toward 3–4 and mean toward the option premium —premium harvesting, not tighter hedges. A separate liquid 10 bp side study (Section 5) is recipe-specific and brittle and does not reopen the headline.

1. Introduction

A desk sells an at-the-money one-month call and hedges with the underlying at discrete times; each trade pays a proportional cost. More rebalancing tracks the delta but spends more; less rebalancing saves cost but leaves variance.

Sepp (2013) maximises the Sharpe ratio inside an analytic delta-hedging family and maps the optimum to price and delta bands. Buehler et al. (2019) replace that family with a network trained on a convex risk measure of terminal P&L, free to trade less or hedge partially. The usual reading is that extra freedom should dominate once costs bite—but that expectation has never been tested on the same paths, contract and cost schedule with a classical side re-tuned in every cell.

The headline before the details. On our frozen 20 bp benchmark (Heston, volatility risk premium, classical band re-tuned per map cell), learned minus classical Sharpe edges run from about -0.09 to $+0.01$; jump models favour classical. A locked holdout on five unseen seeds gives mean edge $+0.0339$ (minimum $+0.0293$, win rate 1.00)—real but tiny. Where a small Heston edge survives, it is not free: terminal P&L spread is almost unchanged (std 0.626 vs 0.613) but skew is -0.70 against -0.01 and CVaR at 95% is 1.321 against 1.066, a 24% worse left tail. Classic hedging is all you need at nominal friction; deep hedging is not a uniform upgrade.

The twist. Above roughly 50 bp the network posts large Sharpe wins, yet its P&L standard deviation climbs from about 0.5 toward 3–4 while mean P&L approaches the option premium—the signature of premium harvesting, not better hedging. Section 4 delivers that comparison after Section 3 checks that the simulator and both papers' implementations are trustworthy.

2. A fair head-to-head

One testbed. Four Sepp market models (GBM, jump-diffusion, Heston, Heston with jumps), one at-the-money one-month call, one proportional cost c on every hedge trade including open and close. The head-to-head map spans four models and five cost levels; the classical baseline is re-tuned in every cell so a stale band cannot manufacture a network win.

Two implementations, one rollout. `hedging/sepp.py` supplies the classical family; `hedging/deep.py` is an MLP on log-moneyness, time and position with CVaR loss at $\alpha = 0.65$. Both use the same P&L rollout and cost accounting.

Controls. Disjoint train, validation, tuning, search and holdout seed pools; reported means average three training seeds with per-seed minima and win rates published. The frozen benchmark adds a 25% implied versus 20% realised volatility risk premium so "do not hedge" cannot dominate Sharpe. The frictionless check is the correctness anchor: at zero cost the learned policy must recover the Black-Scholes delta.

3. Replication checks

3.1 Sepp (2013)

The closed forms track Table 2. The largest relative error in u is 3.9%, and the GBM case I pair (0.5693 against 0.5692) and the SVJ case I pair (0.6551 against 0.6550) differ by 0.0001. The optimum is interior and moves to lower frequency as the model gains complexity: extra trading cannot remove a variance floor f, so paying more cost buys less. GBM case I optimises at 242 rebalances with a Sharpe of 0.83; Heston case I at 75 with 0.43.

The jump-diffusion row needed a real fix. The JD branch had dropped a factor $q = \sqrt{\pi^3}/4$ that the GBM and SVJ branches carry. `archive/scripts/diagnose_jd.py` shows the second-moment gamma was correct to 0.2% all along; only the coefficient was wrong. Restoring q moves p from 24% below the paper to 5.6% above it and brings the analytic JD Sharpe (0.65) onto the simulated one (0.65) in case I. The remaining case-II gap of about 12% in volatility is Sepp's own variance approximation. The Monte Carlo cross-check passes for the other three models in both mean and volatility.

3.2 Deep hedging (Buehler et al. 2019)

At zero cost the learned policy rediscovers the delta. Over 20,000 paths the correlation between learned position and Black-Scholes delta is 0.9907, with slope 1.148 and RMSE 0.0581; the mean learned position (0.499) is within 2% of the mean delta (0.508).

The Heston experiment then reproduces the paper's qualitative claim. At zero cost the network matches a delta hedge (Sharpe 1.154 against 1.139). As cost rises it moves ahead of both every-step delta hedging and the frozen band.

Cost (bp)	Learned	BS delta	Frozen band	Edge vs delta	Edge vs band
0	1.154	1.139	0.987	+0.016	+0.167
20	0.373	0.254	0.352	+0.119	+0.020
50	0.105	−0.970	−0.526	+1.075	+0.631
100	0.182	−2.276	−1.573	+2.458	+1.755

Table 1: Heston Sharpe by cost. The learned hedger pulls ahead as cost rises; Section 4 pairs Sharpe with P&L spread.

4. Head-to-head comparison

The edge map has three regimes, and the P&L standard deviations explain the third.

Near zero cost the two approaches tie, which is the expected outcome once the frictionless check passes. At 20 bp—the benchmark's own cost—the comparison is essentially flat: edges run from −0.09 to +0.01, and on the jump models the learned hedger is slightly behind. This is the headline result: classic hedging is all you need at nominal friction. Above 50 bp the network's Sharpe edge becomes large and consistent, but its P&L standard deviation climbs from about 0.5 to 3 or 4 while its mean P&L approaches the option premium. The network has learned to stop hedging and collect the volatility risk premium. That is a legitimate strategy but a different product from a hedge; a Sharpe win produced by not hedging is a change of risk profile, not evidence of better hedging.

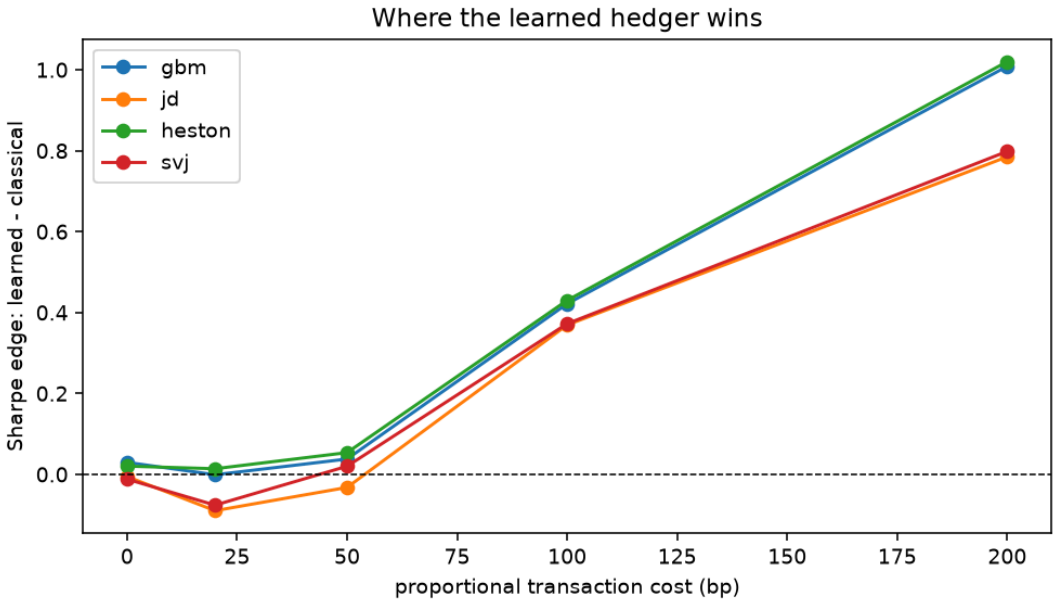


Figure 1: Learned minus classical Sharpe. High-cost wins arrive with a tripled P&L spread.

The locked holdout confirms the moderate-cost result and its size. On five unseen seeds the mean edge over the frozen band is +0.0339, with a minimum of +0.0293 and a win rate of 1.00—the same order of magnitude as the validation edge for Heston at 20 bp (+0.014 fixed hyperparameters, +0.021 after search). The small benchmark win is real but small by design: this is the regime where the two strategies nearly tie.

The Sharpe edge is not a tail win. On every one of the five holdout seeds the learned CVaR95 is higher than the band, from 1.273 to 1.318 against 1.036 to 1.080.

It is also not a free improvement. On the validation distribution the two P&L distributions have almost the same spread, but the learned one has skew −0.70 against −0.01 and a CVaR at 95% of 1.321 against 1.066, which is a 24% worse tail. Whether the edge counts as a win depends on whether the desk scores Sharpe or tail risk, and the two measures disagree.

4.1 Mechanism checks (not the headline)

CVaR tail fraction, state ablations and a participation dial in an extended classical family show that high-cost Sharpe gaps track under-hedging and the map's rule that classical strategies always open a full delta at $t = 0$. Details are in Appendix A figures; they do not overturn the 20 bp tie.

5. Footnote: liquid 10 bp full-rate

Sepp Case I charges $\frac{k}{2}$ on $|\Delta\delta| \cdot S$ per eq. (2.5); our simulator uses a full-rate ledger `cost · |trade| · S`. At Sepp liquid-index k the side study sets `cost = 0.001` (numerically Sepp's k, not Case I economics) and re-sweeps a 10 bp classical hull—never the frozen 20 bp `classical_best.json`—asking whether any deep recipe clears the validation-Sharpe champion on both cost and tail. The chain is soft and mandate-specific: H clears on narrow margins, K is dominated by default training, L clears only at selected $\alpha = 0.05$ seeds, I fails pooled robustness, and J's clearing recipes look band-like. It does not reopen the 20 bp headline.

6. Limitations

This is a course-scale testbed: 80 to 100 epochs, 2,048 training paths per seed, three seeds, a small network and eight Optuna trials per cell. The direction of the results is stable across fixed and searched configurations, but the exact edges are not paper-grade magnitudes. The classical family in the map always opens a full initial hedge, so it cannot express "under-hedge" or "hold flat"; the sidequest repairs

that for interpretation but stayed outside the frozen benchmark. The comparison scripts are covered by smoke tests rather than the unit suite, so their results rest on seed discipline.

7. Conclusion

Classic hedging is all you need at the course's 20 bp benchmark. The trained network ties a well-tuned classical delta band on Sharpe (edges about -0.09 to $+0.01$), and the holdout confirms only a tiny gap. Where a small Heston edge survives, it buys a worse left tail in CVaR. High-cost Sharpe advantages are premium harvesting—visible in the paired P&L standard deviations—not better hedges, and deep is not a uniform upgrade across the comparison grid. The replications stand: Sepp's coefficients match Table 2, closed forms agree with Monte Carlo, and the frictionless network recovers the Black-Scholes delta. The liquid 10 bp footnote (Section 5) does not change that conclusion.

References

[1] H. Buehler, L. Gonon, J. Teichmann and B. Wood. Deep hedging. *Quantitative Finance*, 19(8):1271-1291, 2019. doi:10.1080/14697688.2019.1571683.

[2] A. Sepp. When you hedge discretely: Optimization of Sharpe ratio for delta-hedging strategy under discrete hedging and transaction costs. *Journal of Investment Strategies*, 3(1):19-59, 2013.

Appendix A. Extended tables

A.1 Sepp Table 2 reproduction

Case	Model	u ours	u paper	p ours	p paper	N* ours	N* paper	S ours	S paper
I	GBM	0.5693	0.5692	0.2519	0.2461	242	239	0.83	0.89
I	JD	0.5640	0.5510	0.2118	0.2006	147	138	0.65	0.66
I	Heston	0.6621	0.6703	0.2583	0.2524	75	74	0.43	0.45
I	SVJ	0.6551	0.6550	0.2164	0.2085	70	68	0.43	0.45
II	GBM	1.8080	1.8558	0.6138	0.5998	130	129	1.18	1.30
II	JD	1.7826	1.7628	0.4765	0.4423	88	75	1.01	0.99
II	Heston	2.0054	2.0876	0.6359	0.6214	49	49	0.65	0.73
II	SVJ	1.9738	1.9995	0.4903	0.4634	45	43	0.68	0.72

Table 2: Closed-form coefficients against Sepp's Table 2. Largest relative error in u is 3.9%.

A.2 Monte Carlo cross-check at the analytic optimum

Case	Model	P&L analytic	P&L MC	vol analytic	vol MC	Sharpe a/MC
I	GBM	0.00318	0.00309	0.00385	0.00365	0.83 / 0.85
I	JD	0.00373	0.00381	0.00571	0.00590	0.65 / 0.65
I	Heston	0.00511	0.00537	0.01191	0.01173	0.43 / 0.46
I	SVJ	0.00515	0.00542	0.01187	0.01168	0.43 / 0.46
II	GBM	0.01098	0.01102	0.00932	0.00916	1.18 / 1.20
II	JD	0.01220	0.01295	0.01213	0.01370	1.01 / 0.95
II	Heston	0.01526	0.01695	0.02338	0.02369	0.65 / 0.72
II	SVJ	0.01539	0.01709	0.02280	0.02380	0.68 / 0.72

Table 3: 60,000-path cross-check. Mean matches for every model; the residual case-II volatility gap is the paper's own approximation.

A.3 Full edge map, fixed configuration

Model	Cost	Edge	Edge min	Win	Deep Sharpe	Classical Sharpe	Deep std / Cls std
GBM	0 bp	+0.030	+0.014	1.00	1.258	1.228	0.46 / 0.47
GBM	20 bp	−0.001	−0.008	0.50	0.347	0.348	0.55 / 0.58
GBM	50 bp	+0.038	+0.032	1.00	0.073	0.035	1.72 / 1.74
GBM	100 bp	+0.422	+0.416	1.00	0.161	−0.261	3.44 / 1.74
GBM	200 bp	+1.009	+1.004	1.00	0.158	−0.851	3.48 / 1.74
JD	0 bp	−0.006	−0.009	0.00	0.116	0.121	1.98 / 1.89
JD	20 bp	−0.090	−0.096	0.00	−0.081	0.008	1.94 / 2.47
JD	50 bp	−0.032	−0.037	0.00	−0.149	−0.117	2.19 / 2.47
JD	100 bp	+0.369	+0.361	1.00	0.044	−0.325	4.21 / 2.47
JD	200 bp	+0.785	+0.770	1.00	0.042	−0.743	4.37 / 2.46
Heston	0 bp	+0.020	+0.007	1.00	1.159	1.139	0.51 / 0.52

Model	Cost	Edge	Edge min	Win	Deep Sharpe	Classical Sharpe	Deep std / Cls std
Heston	20 bp	+0.014	+0.006	1.00	0.366	0.352	0.61 / 0.61
Heston	50 bp	+0.054	+0.047	1.00	0.100	0.046	1.60 / 1.74
Heston	100 bp	+0.430	+0.425	1.00	0.179	−0.251	3.11 / 1.73
Heston	200 bp	+1.020	+1.015	1.00	0.174	−0.846	3.17 / 1.73
SVJ	0 bp	−0.012	−0.016	0.00	0.113	0.125	2.11 / 1.92
SVJ	20 bp	−0.076	−0.082	0.00	−0.062	0.014	2.05 / 2.45
SVJ	50 bp	+0.020	+0.016	1.00	−0.092	−0.112	2.36 / 2.45
SVJ	100 bp	+0.372	+0.368	1.00	0.050	−0.322	3.97 / 2.45
SVJ	200 bp	+0.798	+0.789	1.00	0.054	−0.744	4.10 / 2.44

Table 4: Learned minus classical Sharpe. The standard-deviation columns are the guard rail: where the edge is large, the network has stopped hedging.

A.4 Optuna hyperparameter search

Model	Cost	Search edge	Reported edge	Win	Deep Sharpe	Classical Sharpe	Deep std / Cls std
Heston	20 bp	+0.011	+0.021	1.00	0.374	0.352	0.60 / 0.61
Heston	50 bp	+0.140	+0.139	1.00	0.185	0.046	3.09 / 1.74
Heston	100 bp	+0.436	+0.435	1.00	0.185	−0.251	3.18 / 1.73
JD	20 bp	−0.075	−0.077	0.00	−0.069	0.008	1.96 / 2.47
JD	50 bp	+0.158	+0.156	1.00	0.039	−0.117	3.98 / 2.47
JD	100 bp	+0.379	+0.377	1.00	0.052	−0.325	4.33 / 2.47

Table 5: Optuna search ranked on a disjoint search pool. The search does not change the qualitative picture.

A.5 Terminal P&L distribution on the frozen benchmark

Strategy	Mean	Std	Skew	CVaR 95%	5% quantile
Learned	0.240	0.626	−0.699	1.321	−0.880
Band	0.216	0.613	−0.010	1.066	−0.752

Table 6: Same spread, worse left tail. The +0.03 Sharpe edge is partly bought with a 24% worse CVaR.

A.6 Sidequest participation frontier

Model	Cost	Tuned lambda	Freq	Participation Sharpe	Learned Sharpe	Learned - part.	No-hedge Sharpe
GBM	0 bp	1.00	1	+1.228	+1.258	+0.030	+0.165
GBM	20 bp	1.00	3	+0.308	+0.347	+0.039	+0.165
GBM	100 bp	0.00	1	+0.165	+0.161	−0.004	+0.165
GBM	200 bp	0.00	1	+0.165	+0.158	−0.008	+0.165
JD	20 bp	0.00	1	+0.053	−0.081	−0.135	+0.053
JD	100 bp	0.00	1	+0.053	+0.044	−0.009	+0.053

Model	Cost	Tuned lambda	Freq	Participation Sharpe	Learned Sharpe	Learned - part.	No-hedge Sharpe
Heston	0 bp	1.00	1	+1.139	+1.159	+0.020	+0.187
Heston	20 bp	1.00	3	+0.317	+0.366	+0.049	+0.187
Heston	100 bp	0.00	1	+0.187	+0.179	−0.007	+0.187
Heston	200 bp	0.00	1	+0.187	+0.174	−0.012	+0.187
SVJ	20 bp	0.00	1	+0.060	−0.062	−0.122	+0.060
SVJ	100 bp	0.00	1	+0.060	+0.050	−0.010	+0.060
SVJ	200 bp	0.00	1	+0.060	+0.054	−0.006	+0.060

Table 7: The extended classical family (participation dial lambda). Where the network posts a high-cost Sharpe win, the tuned dial is zero and the network is below do-nothing.

A.7 Figures

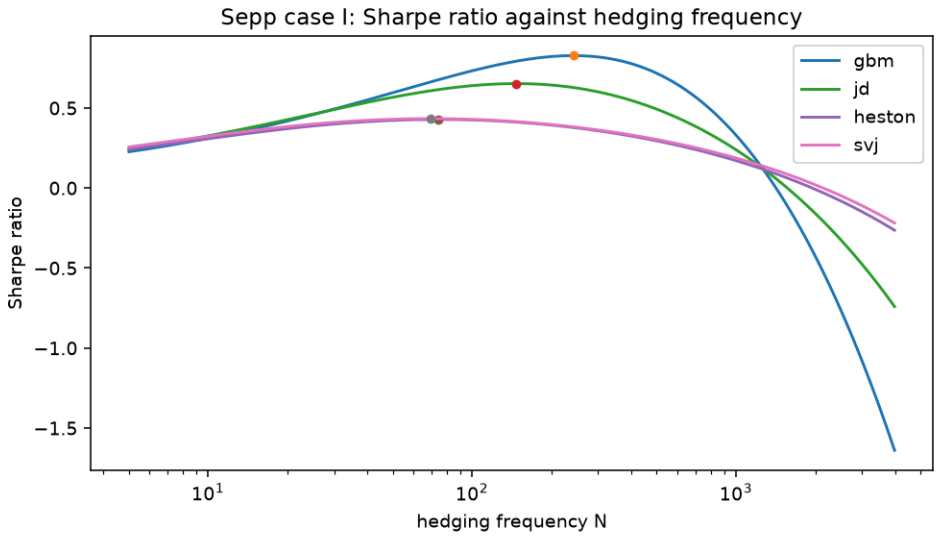


Figure 2: Sepp case I: Sharpe against hedging frequency for the four models. The optimum is interior once cost is positive.

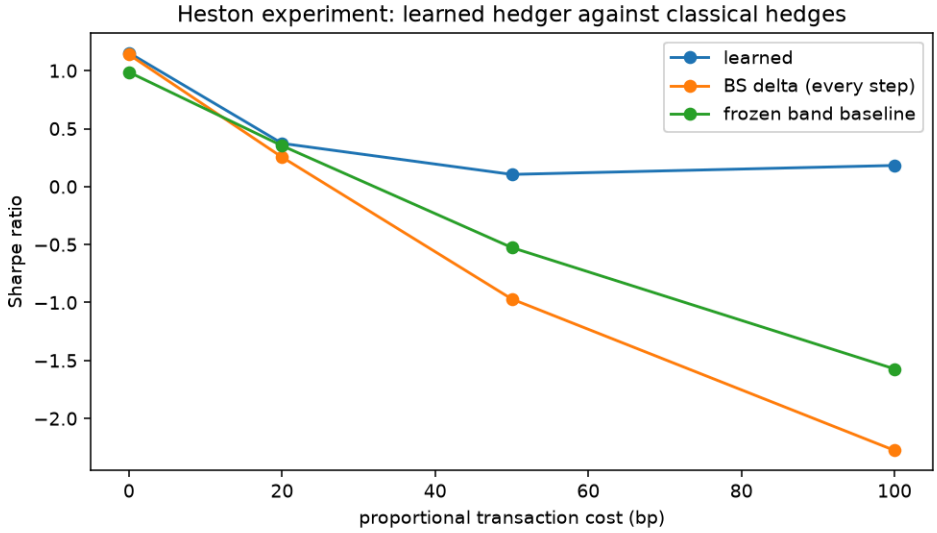


Figure 3: Heston deep-hedging replication: Sharpe against transaction cost for the learned, BS delta and band strategies.

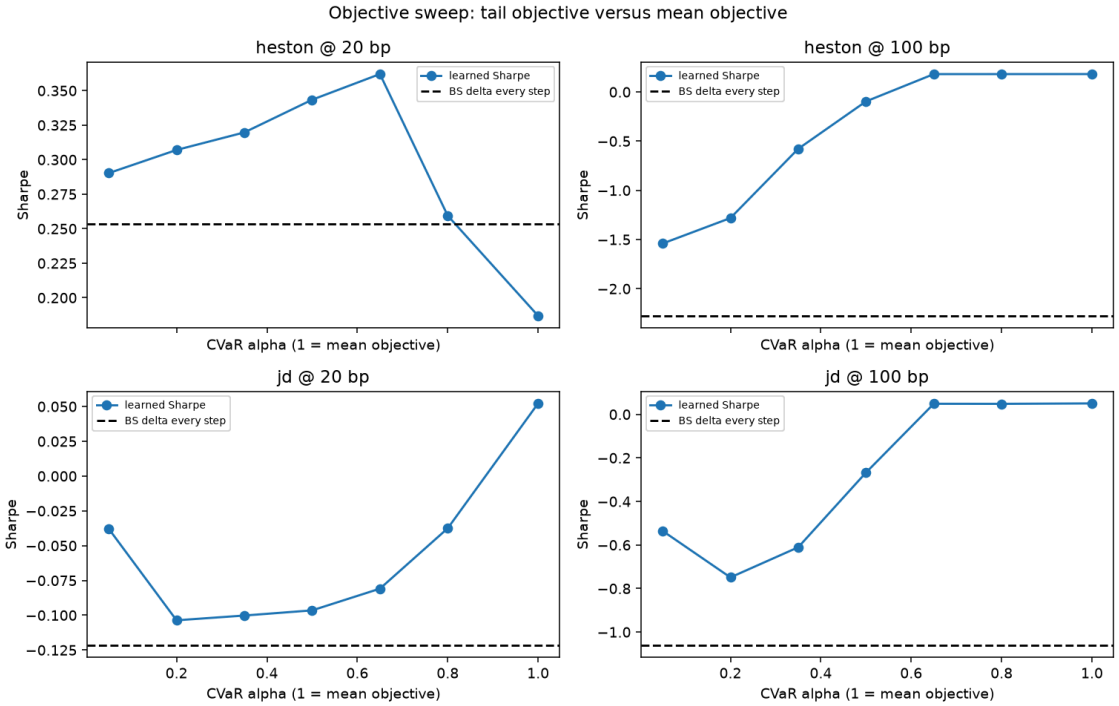


Figure 4: CVaR tail fraction as a turnover dial. Large α collapses trading frequency and inflates P&L spread.

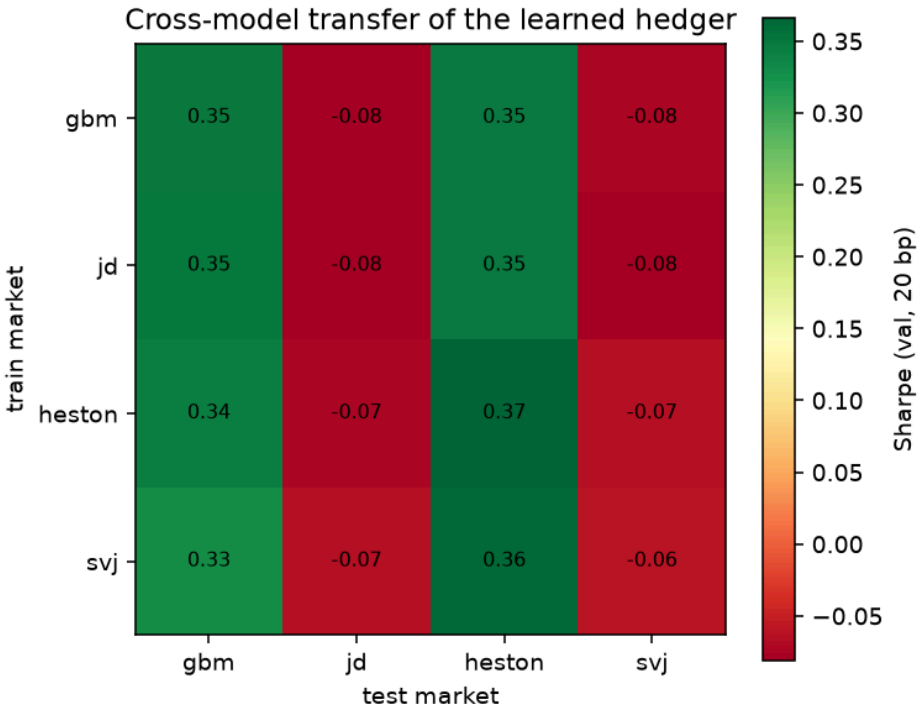


Figure 5: Cross-model transfer: training on one model and scoring on all four. The column order dominates, so the ranking is a property of the market.

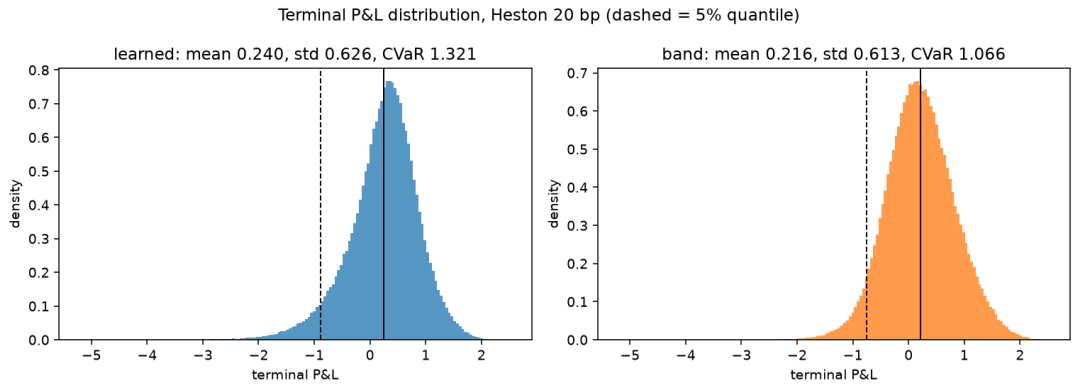


Figure 6: Terminal P&L on the frozen benchmark. The learned distribution has the same spread as the band but a heavier left tail.

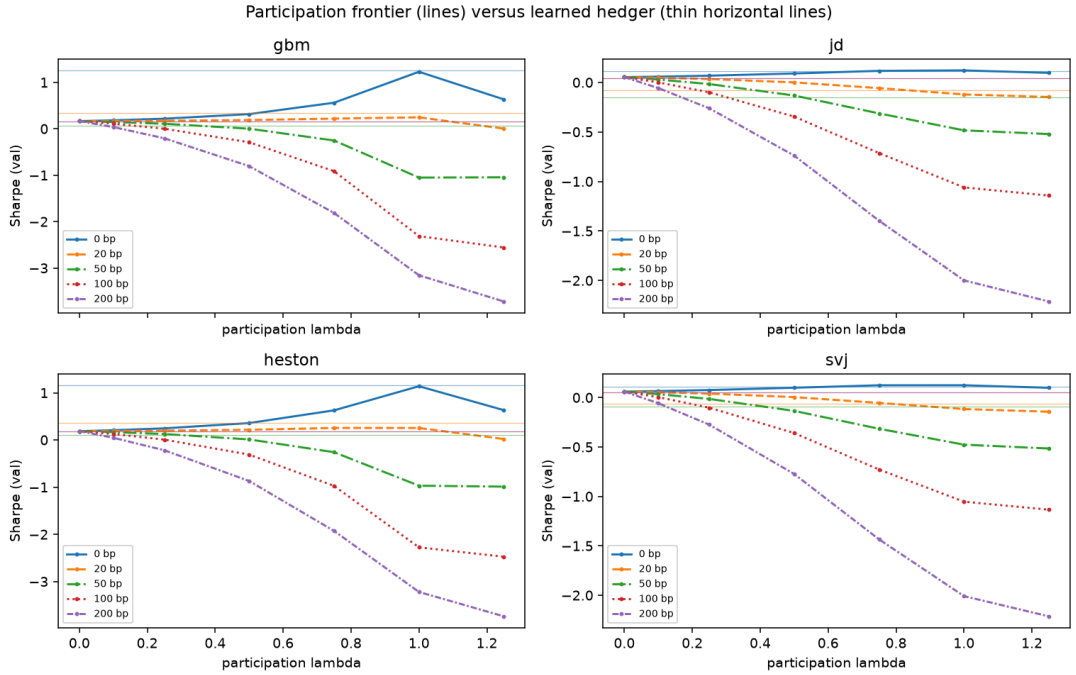


Figure 7: Participation frontier. Above 20 bp the tuned dial is zero and the network sits below do-nothing.

Appendix B. Source code repository

The complete source code is on GitHub:

`github.com/hluaguo/dsc4107`

The repository holds every module used to produce the numbers in this report: the four market simulators, the shared P&L rollout, the Sepp closed forms, the learned hedger and the comparison and exploration scripts. It also carries the test suite and the pinned environment. A snapshot is attached under `submission/code/`, and the raw experimental data sits under `submission/data/`.

Path	Role
hedging/market.py	GBM, JD, Heston and SVJ simulators (frozen)
hedging/bs.py	Black-Scholes price and greeks (frozen)
hedging/hedge.py	Shared P&L rollout and classical strategies (frozen)
hedging/metrics.py	Sharpe, CVaR and the reported statistics (frozen)
hedging/bench.py	Frozen benchmark config and seed pools
hedging/sepp.py	Sepp (2013) closed forms and Sharpe optimisation
hedging/deep.py	Learned hedger: features, network, CVaR loss
hedging/hyper.py	Training hyperparameters
scripts/sepp_replication.py	Sepp Table 2, optima and Monte Carlo check
scripts/deep_hedging_replication.py	Frictionless check and Heston experiment
archive/scripts/comparison_study.py	Head-to-head map and Optuna add-on
scripts/holdout_eval.py	The single locked holdout run
`archive/scripts/explore_*.py`	Objective, feature, transfer and robustness probes
archive/scripts/sidequest_participation.py	The participation-dial test
tests/	Analytic and accounting checks (24 tests)

Table 8: Repository layout.

Appendix C. How to reproduce

The full environment is pinned in `uv.lock`. Clone the repository and sync it:

```
# from the repository root
git clone https://github.com/hluaguo/dsc4107.git
cd dsc4107
uv sync
uv run pytest -q                                          # 24 checks
uv run python scripts/sepp_replication.py                 # Sepp tables + figures
uv run python scripts/sepp_replication.py --mc --mc-paths 60000
uv run python scripts/deep_hedging_replication.py --epochs 100 --hidden 64 --depth 3
uv run python archive/scripts/comparison_study.py --fixed-hyper \
    --costs 0.0 0.002 0.005 0.01 0.02
uv run python archive/scripts/comparison_study.py --models heston jd \
    --costs 0.002 0.005 0.01 --trials 8
uv run python scripts/holdout_eval.py                     # final locked run
```

All training scripts select CUDA when it is present and fall back to the CPU otherwise. Raw experimental data and figures are mirrored under `submission/data/`.